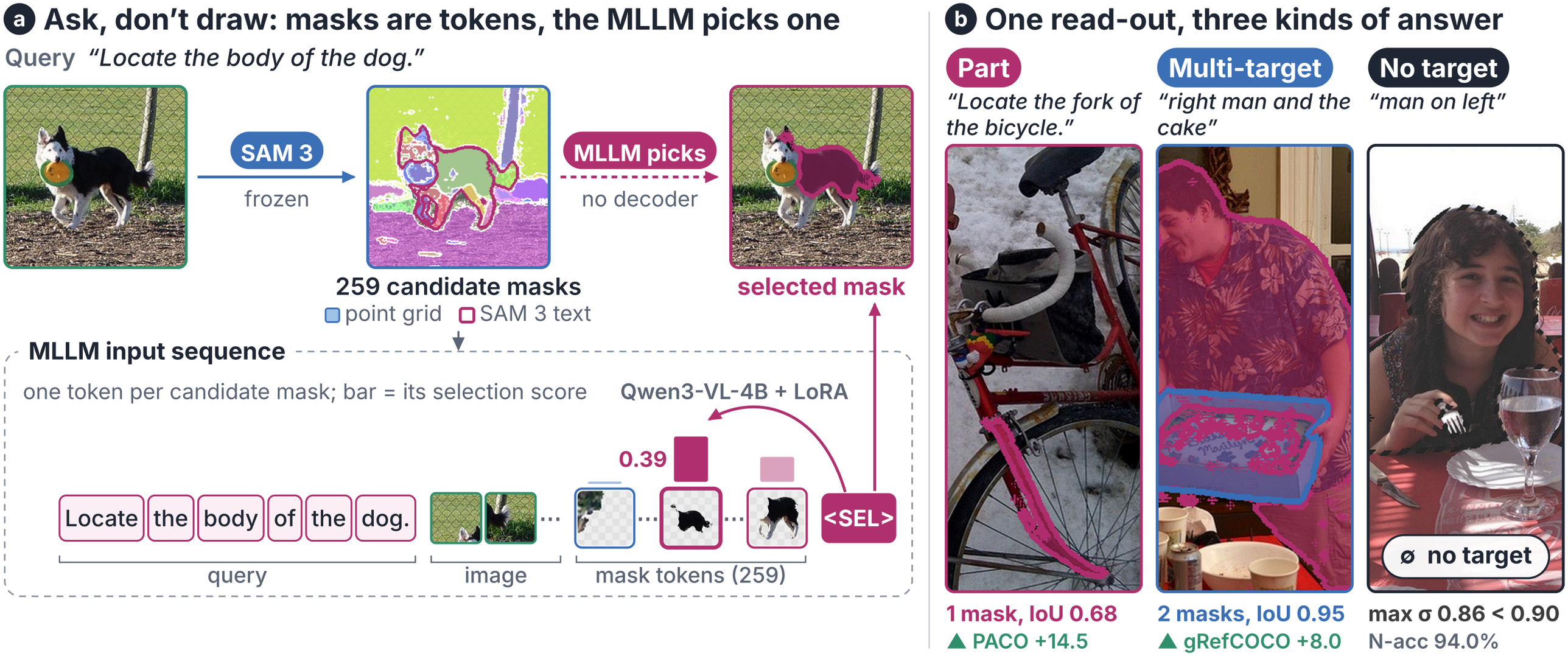

a Ask, don't draw: masks are tokens, the MLLM picks one
Query "Locate the body of the dog."
SAM 3
frozen
259 candidate masks
point grid
SAM 3 text
MLLM picks
no decoder
selected mask
MLLM input sequence
one token per candidate mask; bar = its selection score
Qwen3-VL-4B + LoRA
0.39
Locate
the
body
of
the
dog.
<SEL>
query
image
mask tokens (259)
b One read-out, three kinds of answer
Part
"Locate the fork of the bicycle."
Multi-target
"right man and the cake"
No target
"man on left"
ø no target
1 mask, IoU 0.68
▲ PACO +14.5
2 masks, IoU 0.95
▲ gRefCOCO +8.0
max σ 0.86 < 0.90
N-acc 94.0%